

WASP-52b

R-1542

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_191022 · created 2026-08-01T19:14:06+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458778.8001 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.209 +/- 0.062
Transit depth (Rp/Rs)^2	4.39 +/- 2.62 [%]
Orbital Inclination (inc)	88.9 +/- 2.78
Airmass coefficient 1 (a1)	1.57 +/- 0.98
Airmass coefficient 2 (a2)	-0.28 +/- 0.19
Scatter in the residuals of the lightcurve fit is	65.33 %
Best Comparison Star	#3 - [289, 434]
Optimal Aperture	2.05
Optimal Annulus	3.0
Transit Duration (day)	0.0866 +/- 0.0096

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.209 ± 0.062 vs 0.1646 (deviation 0.6σ)
Duration fit vs published	0.0866 d vs 0.0758 d (deviation 0.93σ)
Mid-time O-C	-11.8 min
Depth SNR	2.8
Residual scatter	65.33 %

Light curve

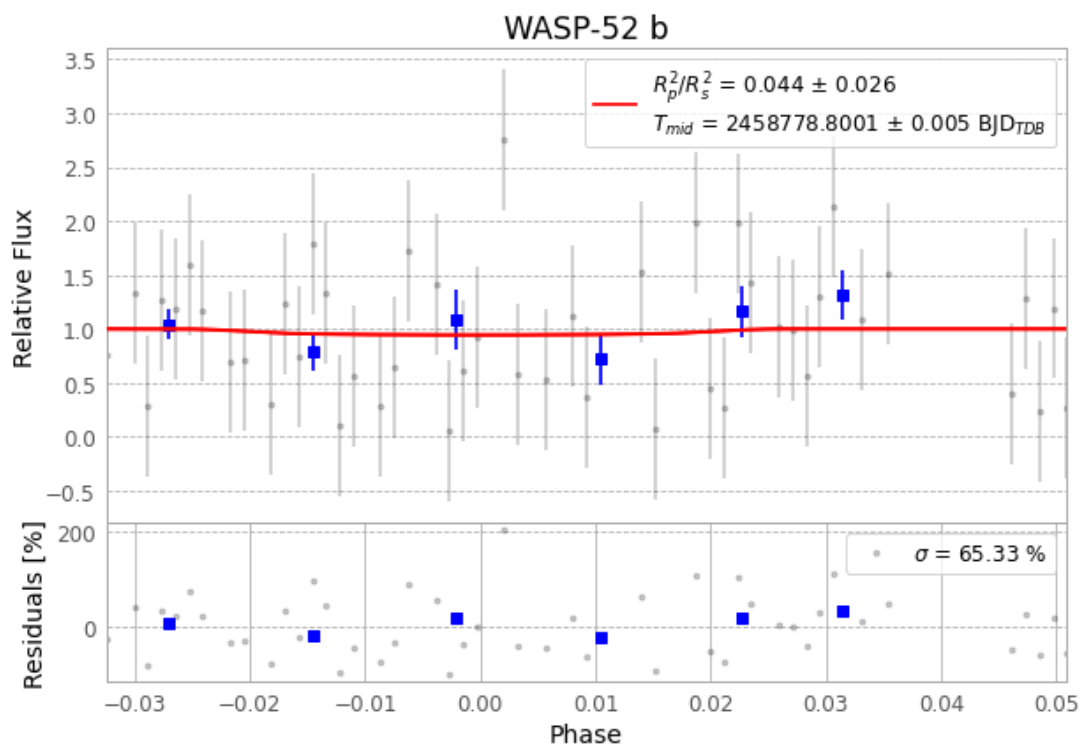


Figure 1 — FinalLightCurve_WASP-52 b_2019-10-21.png (SHA-256 76138cf25df375f1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [489, 415], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b58dc2bfb36e6f73...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

70 FITS frames (46 MB), WASP-52191022054212.FITS → WASP-52191022091512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-191022.log (7558f6495fe1...), FinalLightCurve_WASP-52 b_2019-10-21.png (76138cf25df3...), FinalLightCurve_WASP-52 b_2019-10-21.csv (6f884534417f...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 19:14Z.